

Research Notes on Overfitting machine learning

Overfitting machine learning

>> Section 1

Usually, a learning algorithm is trained using some set of "training data": exemplary situations for which the desired output is known. The goal is that the algorithm will also perform well on predicting the output when fed "validation data" that was not encountered during its training. Overfitting is the use of models or procedures that violate Occam's razor, for example by including more adjustable parameters than are ultimately optimal, or by using a more complicated approach than is ultimately optimal. For an example where there are too many adjustable parameters, consider a dataset where training data for y can be adequately predicted by a linear function of two independent variables. Such a function requires only three parameters (the intercept and two slopes). Replacing this simple function with a new, more complex quadratic function, or with a new, more complex linear function on more than two independent variables, carries a risk: Occam's razor implies that any given complex function is a priori less probable than any given simple function. If the new, more complicated function is selected instead of the simple function, and if there was not a large enough gain in training data fit to offset the complexity increase, then the new complex function "overfits" the data and the complex overfitted function will likely perform worse than the simpler function on validation data outside the training dataset, even though the complex function performed as well, or perhaps even better, on the training dataset. When comparing different types of models, complexity cannot be measured solely by counting how many parameters exist in each model; the expressivity of each parameter must be considered as well. For example, it is nontrivial to directly compare the complexity of a neural net (which can track curvilinear relationships) with m parameters to a regression model with n parameters. Overfitting is especially likely in cases where learning was performed too long or where training examples are rare, causing the learner to adjust to very specific random features of the training data that have no causal relation to the target function. In this process of overfitting, the performance on the training examples still increases while the performance on unseen data becomes worse. As a simple example, consider a database of retail purchases that includes the item bought, the purchaser, and the date and time of purchase. It's easy to construct a model that will fit the training set perfectly by using the date and time of purchase to predict the other attributes, but this model will not generalize at all to new data because those past times will never occur again. Generally, a learning algorithm is said to overfit relative to a simpler one if it is more accurate in fitting known data (hindsight) but less accurate in predicting new data (foresight). One can intuitively

understand overfitting from the fact that information from all past experience can be divided into two groups: information that is relevant for the future, and irrelevant information ("noise"). Everything else being equal, the more difficult a criterion is to predict (i.e., the higher its uncertainty), the more noise exists in past information that needs to be ignored. The problem is determining which part to ignore. A learning algorithm that can reduce the risk of fitting noise is called "robust."

>> Section 2

In statistics, an inference is drawn from a statistical model, which has been selected via some procedure. Burnham & Anderson, in their much-cited text on model selection, argue that to avoid overfitting, one should adhere to the "Principle of Parsimony". The authors also state the following.

>> Section 3

A function that is overfitted is likely to request more information about each item in the validation dataset than does the optimal function; gathering this additional unneeded data can be expensive or error-prone, especially if each individual piece of information must be gathered by human observation and manual data entry. A more complex, overfitted function is likely to be less portable than a simple one. At one extreme, a one-variable linear regression is so portable that, if necessary, it could even be done by hand. At the other extreme are models that can be reproduced only by exactly duplicating the original modeler's entire setup, making reuse or scientific reproduction difficult. It may be possible to reconstruct details of individual training instances from an overfitted machine learning model's training set. This may be undesirable if, for example, the training data includes sensitive personally identifiable information (PII). This phenomenon also presents problems in the area of artificial intelligence and copyright, with the developers of some generative deep learning models such as Stable Diffusion and GitHub Copilot being sued for copyright infringement because these models have been found to be capable of reproducing certain copyrighted items from their training data.

>> Section 4

Overfitted models ... are often free of bias in the parameter estimators, but have estimated (and actual) sampling variances that are needlessly large (the precision of the estimators is poor, relative to what could have been accomplished with a more parsimonious model). False treatment effects tend to be identified, and false variables are included with overfitted models. ... A best approximating model is achieved by properly balancing the

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errors of underfitting and overfitting. Overfitting is more likely to be a serious concern when there is little theory available to guide the analysis, in part because then there tend to be a large number of models to select from. The book *Model Selection and Model Averaging* (2008) puts it this way.

>> Section 5

In mathematical modeling, overfitting is the production of an analysis that corresponds too closely or exactly to a particular set of data, and may therefore fail to fit to additional data or predict future observations reliably. An overfitted model is a mathematical model that contains more parameters than can be justified by the data. In the special case of a model that consists of a polynomial function, these parameters represent the degree of a polynomial. The essence of overfitting is to unknowingly extract some of the residual variation (i.e., noise) as if that variation represents the underlying model structure. Underfitting occurs when a mathematical model cannot adequately capture the underlying structure of the data. An under-fitted model is a model that is missing some parameters or terms that would appear in a correctly specified model. Underfitting would occur, for example, when fitting a linear model to nonlinear data. Such a model will tend to have poor predictive performance. The possibility of over-fitting exists when the criterion used for selecting the model is not the same as the criterion used to judge the suitability of a model. For example, a model might be selected by maximizing its performance on some set of training data, yet its suitability might be determined by its ability to perform well on unseen data; overfitting occurs when a model begins to "memorize" training data rather than "learning" to generalize from a trend. As an extreme example, if the number of parameters is the same as or greater than the number of observations, then a model can perfectly predict the training data simply by memorizing the data in its entirety. (For an illustration, see Figure 2.) Such a model will typically fail severely when making predictions. Overfitting is related to both the complexity of the chosen model and how well it is optimized during training. A function class that is too large, in a suitable sense, relative to the dataset size is likely to overfit. Even when the fitted model does not have an excessive number of parameters, it is to be expected that the fitted relationship will appear to perform less well on a new dataset than on the dataset used for fitting (a phenomenon sometimes known as shrinkage). In particular, the value of the coefficient of determination will shrink relative to the original data. To lessen the chance or amount of overfitting, several techniques are available (e.g., model comparison, cross-validation, regularization, early stopping, pruning, Bayesian priors, or dropout). The

basis of some techniques is to either (1) explicitly penalize overly complex models or (2) test the model's ability to generalize by evaluating its performance on a set of data not used for training, which is assumed to approximate the typical unseen data that a model will encounter.